

Skills Summary

Product, Quotient, and Power Rules for Logarithms

Use this reference while completing your worksheet or when studying.

Skill 1 — The product rule

Rule: $\log_b(MN) = \log_b M + \log_b N$, for the same base b and positive M , N . Read left to right it expands a product; read right to left it condenses a sum into one logarithm.

Example: Give the exact value of $\log_6 4 + \log_6 9$.

Step 1. Neither logarithm is a whole number on its own, but the bases match, so condensing with the product rule may produce a friendly input.

Step 2. $\log_6 4 + \log_6 9 = \log_6(4 \cdot 9) = \log_6 36$, and $6^2 = 36$.
 $\Rightarrow 2$

Try it: Give the exact value of $\log_2 16 + \log_2 4$.

check: 9

Skill 2 — The quotient rule

Rule: $\log_b \frac{M}{N} = \log_b M - \log_b N$. The numerator's logarithm comes first; subtracting in the wrong order gives the negative of the right answer.

Example: Give the exact value of $\log_3 108 - \log_3 4$.

Step 1. A difference of same-base logarithms condenses to one logarithm of a quotient, and dividing first is what turns an awkward input into a power of the base.

Step 2. $\log_3 \frac{108}{4} = \log_3 27$, and $3^3 = 27$.
 $\Rightarrow 3$

Try it: Give the exact value of $\log_5 250 - \log_5 2$.

check: 3

Skill 3 — The power rule

Rule: $\log_b(M^p) = p \log_b M$. An exponent inside comes out in front; a coefficient in front goes back inside as an exponent. A coefficient of $\frac{1}{2}$ is a square root, and a negative coefficient is a reciprocal.

Example: Expand $\log_b(x^4)$ so no exponent is left inside.

Step 1. The input is a single power, so the power rule alone does the whole job — no product or quotient is involved.

Step 2. The exponent 4 moves out in front of the logarithm.
 $\Rightarrow 4 \log_b x$

Try it: Expand $\log_b(y^7)$.

check: $\log_b(y^7)$

Skill 4 — Using all three together

Order that works: to *condense*, move every coefficient inside with the power rule first, then combine sums into products and differences into quotients. To *expand*, split the product or quotient first, then bring each exponent out.

Example: Condense $2\log_b x + \log_b 3 - \log_b y$.

Step 1. Coefficients must go inside before anything can be combined, because the product and quotient rules only apply to plain logarithms.

Step 2. $2\log_b x = \log_b(x^2)$, so the expression becomes $\log_b(x^2) + \log_b 3 - \log_b y$, which condenses to one logarithm.

$$\Rightarrow \log_b \frac{3x^2}{y}$$

Try it: Condense $\log_b 5 + 3\log_b x$.

check: $\log_b(5x^3)$